



LASSE

An Introduction to Boosting and AdaBoost using Weka



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Most slides from P. Viola and others

Boosting

- Boosting is a “meta” learning scheme that works by combining *weak learners*. It can generate an accurate “ensemble classifier” and easily perform “feature selection”
 - A weak learner need only do better than chance
- Training consists of multiple *boosting rounds*
 - During each boosting round, we select a weak learner that does well on examples that were hard for the previous weak learners
 - “Hardness” is captured by weights attached to training examples

Y. Freund and R. Schapire, [A short introduction to boosting](#), *Journal of Japanese Society for Artificial Intelligence*, 14(5):771-780, September, 1999.

Training

Choose a resolution (say, images with 24 x 24 pixels)

Use a large number of “positive” examples
(faces)



And a large number of “negative” examples,
(anything but a face)

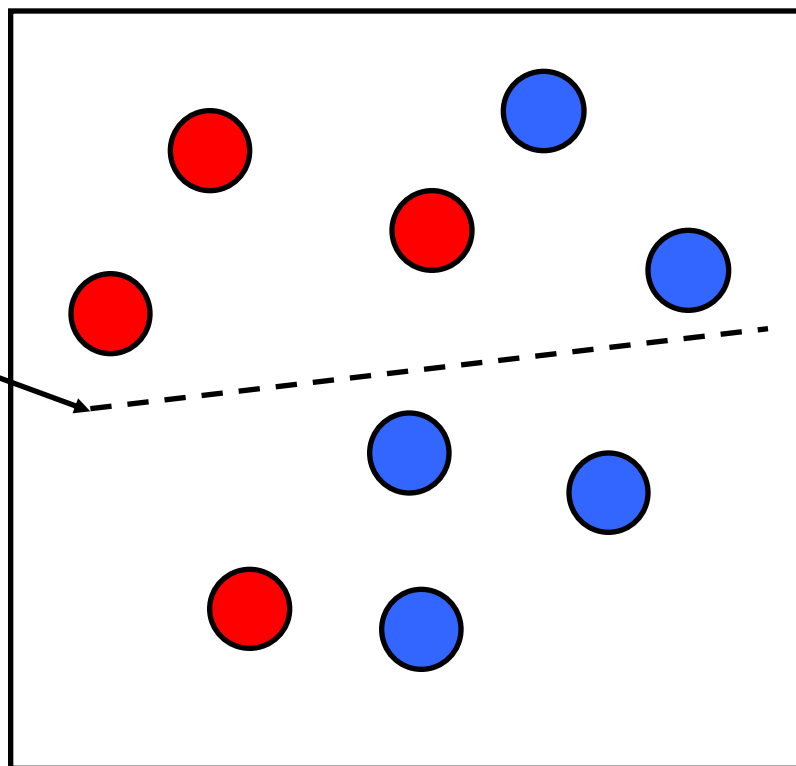
Training procedure

- Initially, weight each training example equally
- In each boosting round:
 - Find the weak learner (we assume it is a Decision Stump) that achieves the lowest *weighted* training error
 - Raise the weights of training examples misclassified by current (partially trained) classifier, taking all weak learners already trained in account
- Compute final classifier as linear combination of all weak learners (weight of each learner is directly proportional to its accuracy)
- Exact formulas for re-weighting and combining weak learners depend on the particular boosting scheme (e.g., AdaBoost)

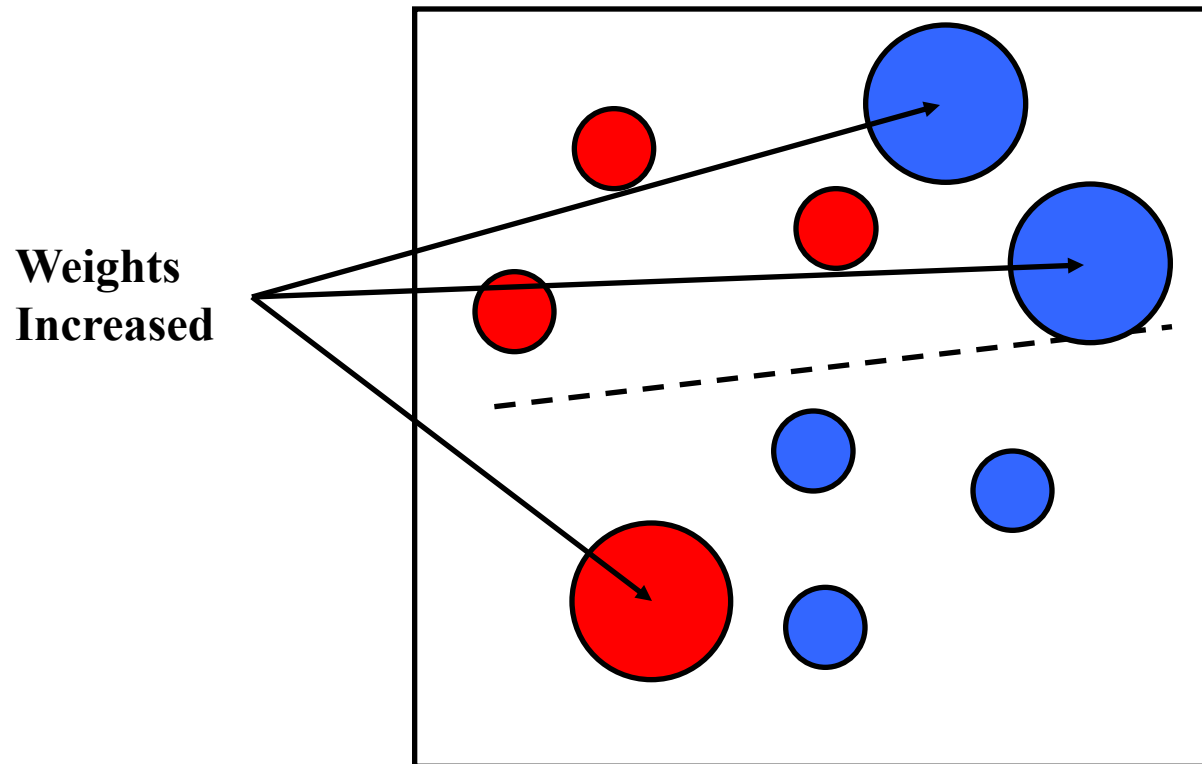
Y. Freund and R. Schapire, [A short introduction to boosting](#), *Journal of Japanese Society for Artificial Intelligence*, 14(5):771-780, September, 1999.

Boosting illustration (perceptron as weak learner)

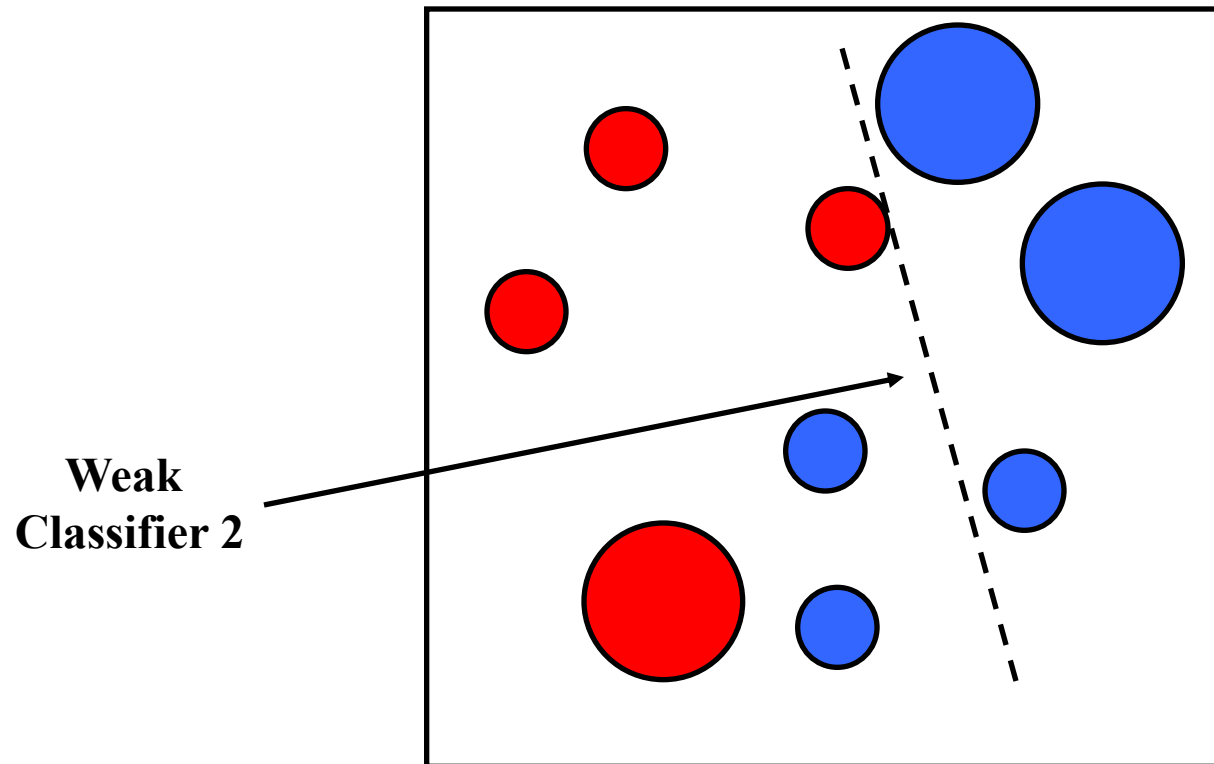
**Weak
Classifier 1**



Boosting illustration

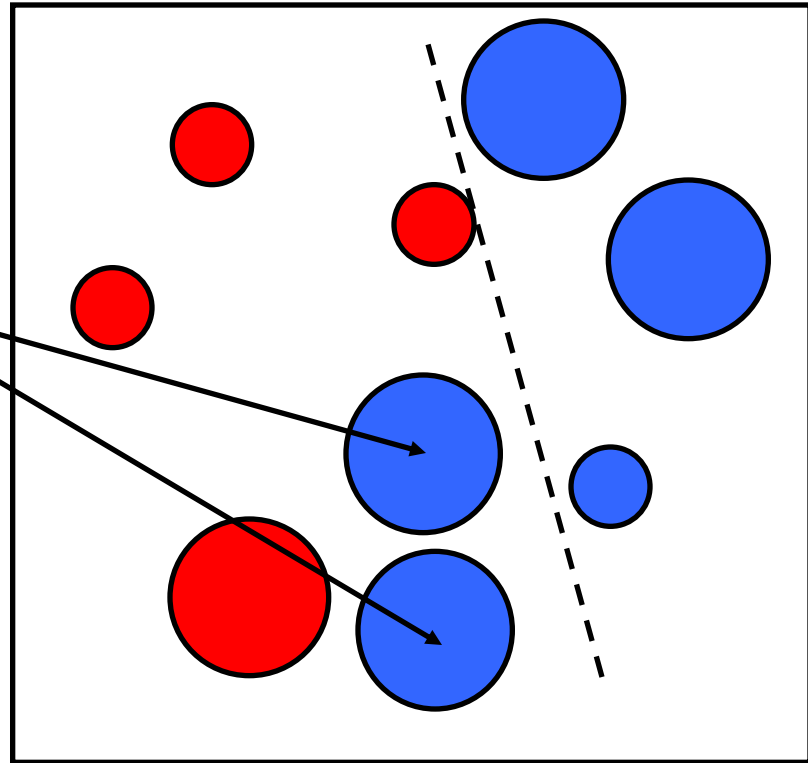


Boosting illustration

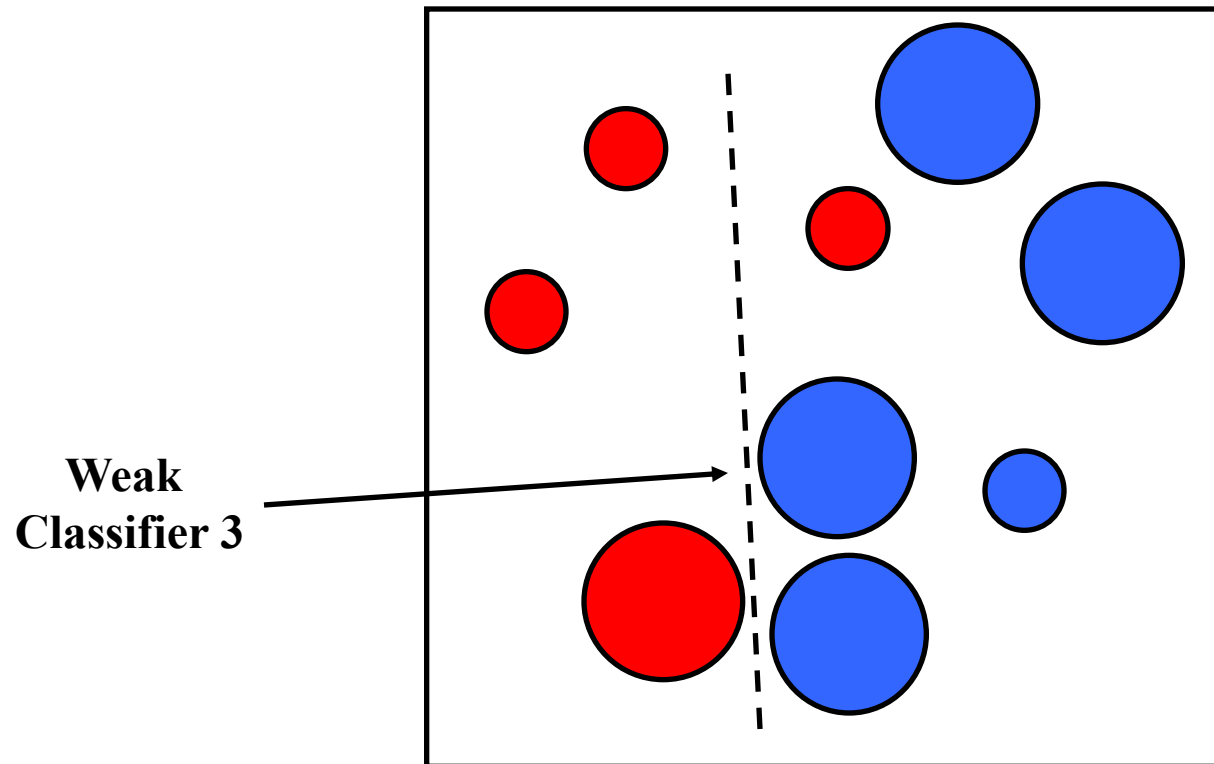


Boosting illustration

**Weights
Increased**

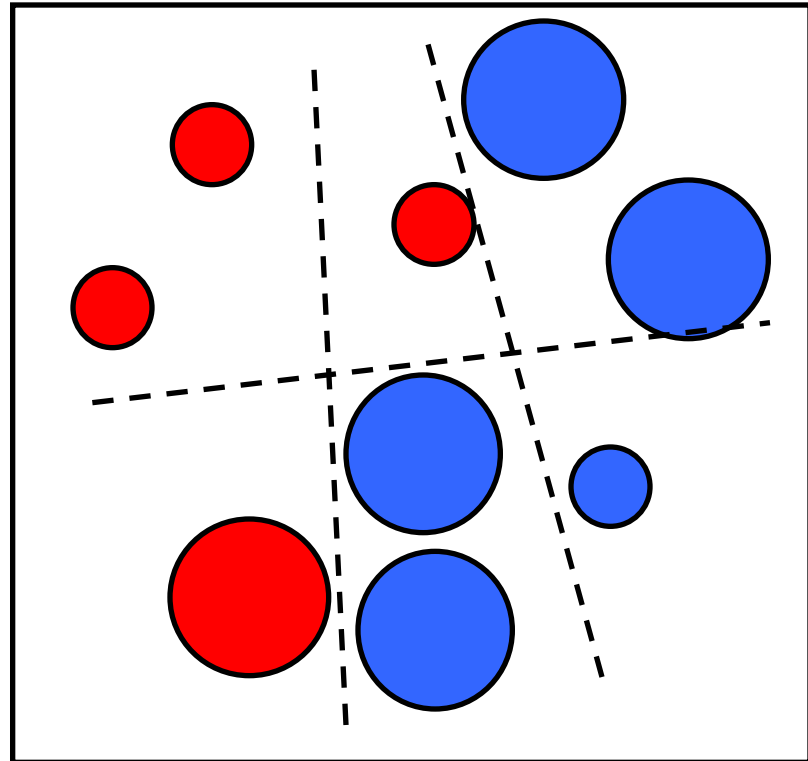


Boosting illustration



Boosting illustration

**Final classifier is
a combination of weak
classifiers**



AdaBoost algorithm

- Given example images $(x_1, y_1), \dots, (x_n, y_n)$ where $y_i = 0, 1$ for negative and positive examples respectively.
- Initialize weights $w_{1,i} = \frac{1}{2m}, \frac{1}{2l}$ for $y_i = 0, 1$ respectively, where m and l are the number of negatives and positives respectively.
- For $t = 1, \dots, T$:

1. Normalize the weights, $w_{t,i} \leftarrow \frac{w_{t,i}}{\sum_{j=1}^n w_{t,j}}$
2. Select the best weak classifier with respect to the weighted error

$$\epsilon_t = \min_{f,p,\theta} \sum_i w_i |h(x_i, f, p, \theta) - y_i|.$$

See Section 3.1 for a discussion of an efficient implementation.

3. Define $h_t(x) = h(x, f_t, p_t, \theta_t)$ where f_t, p_t , and θ_t are the minimizers of ϵ_t .
4. Update the weights:

$$w_{t+1,i} = w_{t,i} \beta_t^{1-e_i}$$

where $e_i = 0$ if example x_i is classified correctly, $e_i = 1$ otherwise, and $\beta_t = \frac{\epsilon_t}{1-\epsilon_t}$.

- The final strong classifier is:

$$C(x) = \begin{cases} 1 & \sum_{t=1}^T \alpha_t h_t(x) \geq \frac{1}{2} \sum_{t=1}^T \alpha_t \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha_t = \log \frac{1}{\beta_t}$

AdaBoost algorithm (cont.)

- Given example images $(x_1, y_1), \dots, (x_n, y_n)$ where $y_i = 0, 1$ for negative and positive examples respectively.
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AdaBoost algorithm (cont. 2)

3. Define $h_t(x) = h(x, f_t, p_t, \theta_t)$ where f_t , p_t , and θ_t are the minimizers of ϵ_t .

4. Update the weights:

$$w_{t+1,i} = w_{t,i} \beta_t^{1-e_i}$$

Case 1:

$$\epsilon_t = 0.3$$

$$\beta_t = 0.43$$

$$\alpha_t = 0.85$$

Case 2:

$$\epsilon_t = 0.1$$

$$\beta_t = 0.11$$

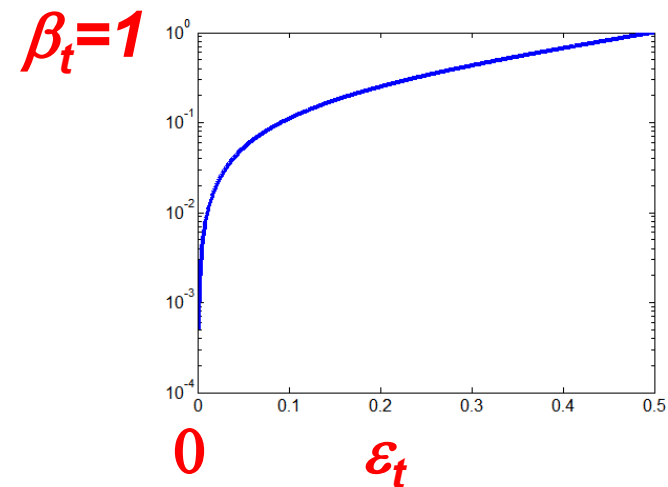
$$\alpha_t = 2.2$$

where $e_i = 0$ if example x_i is classified correctly, $e_i = 1$ otherwise, and $\beta_t = \frac{\epsilon_t}{1-\epsilon_t}$.

• The final strong classifier is:

$$C(x) = \begin{cases} 1 & \sum_{t=1}^T \alpha_t h_t(x) \geq \frac{1}{2} \sum_{t=1}^T \alpha_t \\ 0 & \text{otherwise} \end{cases}$$

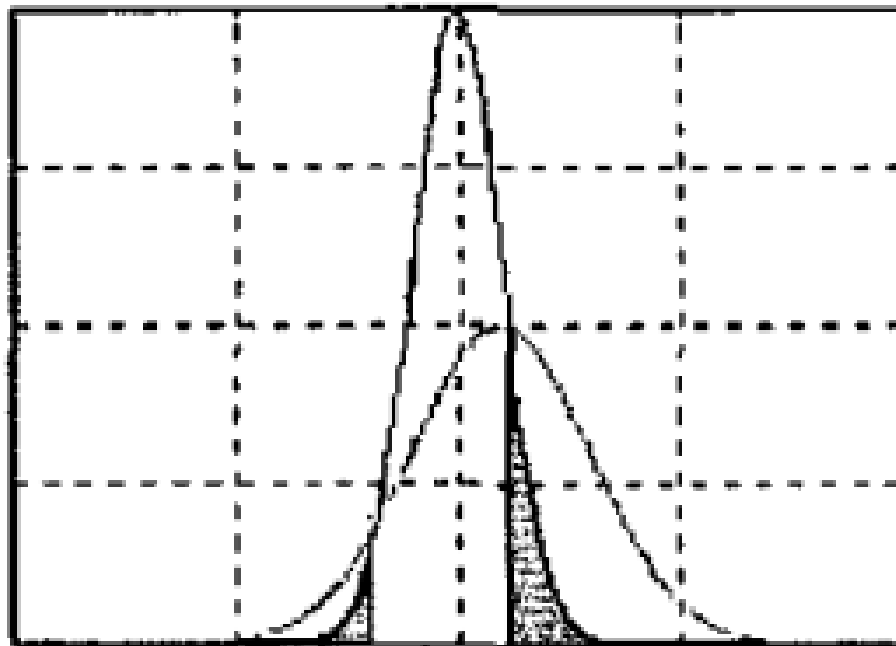
where $\alpha_t = \log \frac{1}{\beta_t}$



Faster boosting for binary problems

“Mining speech: automatic selection of heterogeneous features using boosting”, A. Klautau, ICASSP, 2003 (implemented as Weka class: FastBoostingFeatureSelector.java)

Fit two Gaussians and use the Bayes' error:



There are several boosting algorithms

See a review at

http://cbio.mskcc.org/~aarvey/boosting_papers.html

Some improvements:

- support to multiclass (not only binary)

- support to multilabel

- use confidence-rated predictions (i.e., real numbers instead of an integer)